Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

Abstract

In today's digital environment, organizations rely heavily on structured and accurate data management systems. One of the major challenges faced is importing external data into enterprise platforms without losing relationships between records. This project focuses on automating the process of importing employee data into the ServiceNow platform using Import Sets and Transform Maps.

Import Sets act as staging tables where external data is temporarily stored before being processed. Transform Maps are then used to map this data into target tables while maintaining relationships such as manager and department. Additionally, dot walking is implemented to retrieve related data dynamically in forms like Incident without opening individual records.

The project improves data accuracy, reduces manual effort, and enhances usability. It demonstrates how structured data transformation and relationship mapping can be efficiently handled in ServiceNow.

Introduction

Organizations frequently deal with employee data stored in external sources such as Excel, CSV, or XML files. Importing this data manually into systems like ServiceNow is inefficient and prone to errors. Moreover, maintaining relationships between records such as employee-manager hierarchy and department linkage becomes difficult.

The ServiceNow platform provides tools like Import Sets and Transform Maps to address these challenges. Import Sets serve as temporary storage for incoming data, while Transform Maps define how the data is mapped into target tables.

Problem Statement

- Manual data entry leads to errors and inconsistencies
- Lack of proper relationship mapping affects data integrity
- Difficulty in retrieving related data across tables

Objectives

- Automate data import into ServiceNow
- Maintain relationships between users, departments, and managers
- Use Transform Maps for accurate data mapping
- Implement dot walking for efficient data retrieval
- Improve user experience in Incident forms

Literature Review:

ServiceNow provides a robust framework for data import and transformation. Several concepts are essential for understanding this process:

Import Sets:

Import Sets are temporary tables used to store external data before processing. They act as staging areas and prevent direct impact on production tables.

Transform Maps:

Transform Maps define how data from the import set is transferred to the target table. They include field mappings and transformation logic.

Field Mapping:

Field mapping ensures that each column in the source data corresponds to the correct field in the target table. This can be done automatically or manually using mapping tools.

Dot Walking:

Dot walking is a technique used in ServiceNow to access related fields across tables. It allows users to retrieve information without opening the referenced record.

Existing Systems:

- Manual data entry systems
- Basic import tools without relationship mapping
- Script-based imports requiring technical expertise

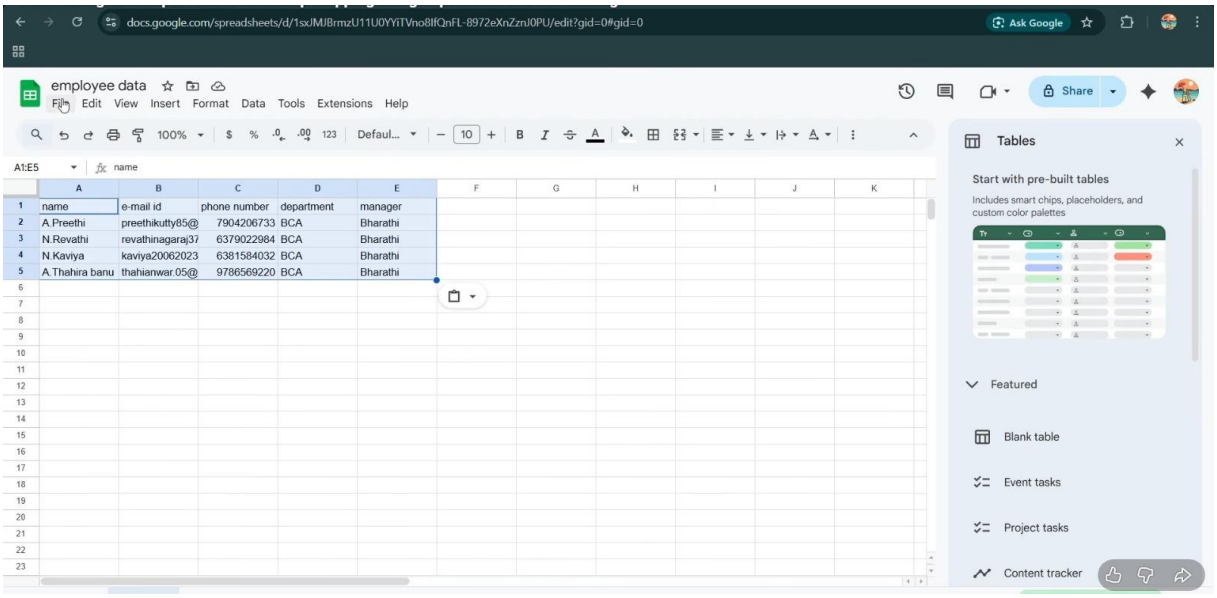
This project improves upon existing methods by combining Import Sets, Transform Maps, and dot walking into a single automated workflow.

Methodology:

The project follows a structured step-by-step methodology:

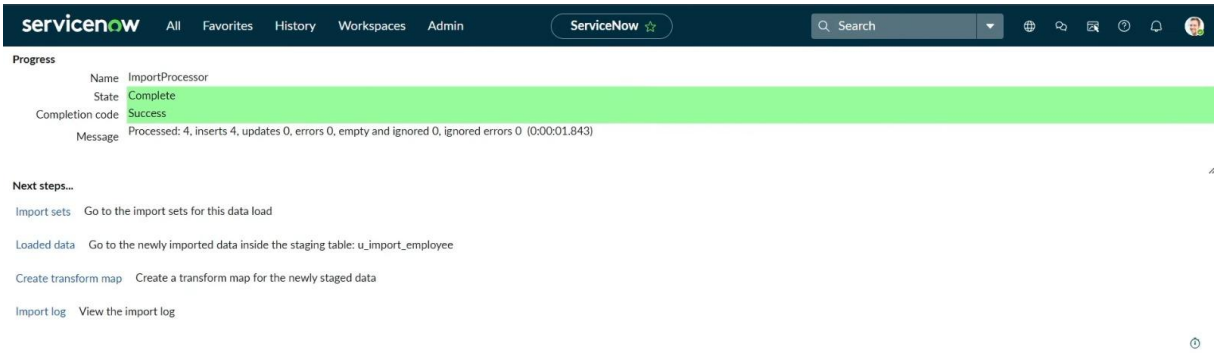
Step 1: Data Preparation

- Create an Excel file containing employee details
- Fields include:
    - Name
    - Email
    - Department
    - Manager
    - Phone Number



Step 2: Load Data into ServiceNow

- Navigate to Import Sets
- Select “Load Data”
- Upload the Excel/CSV/XML file
- Create a new import set table



Step 3: Create Transform Map

- Navigate to “Create Transform Map”
- Provide name (e.g., Test)
- Select:
    - Source Table → Import Set Table
    - Target Table → User [sys_user]

servicenow All Favorites History Workspaces Table Transform Map - New Record Search

Table Transform Map
New record

Submit

* Name: employee transform
* Source table: import employee [u_import_employee]
Active: ☑
Run business rules: ☑
Enforce mandatory fields: No
Copy empty fields: ☐
Create new record on empty coalesce fields: ☐

Application: Global
Created:
* Target table: User [sys_user]
Order: 100
Run script: ☐

Submit

servicenow All Favorites History Workspaces Admin Table Transform Maps Search

Table Transform Maps Order Search Actions on selected r

All

Name	Source table	Target table	Run business rules	Order	Active	Up
Location	Location [imp_location]	Location [cmn_location]	true	100	true	201
Computer	Computer [imp_computer]	Computer [cmdb_ci_computer]	true	100	true	201
ADFS Example	label [oidc_adfs_import]	User [sys_user]	true	100	true	201
Google Example	label [oidc_google_import]	User [sys_user]	true	100	true	201
Employee Job Profile Transform Map	Employee Job Profile Staging [sn_employee_position_stg]	Employee Job Profile [sn_employee_position]	true	100	true	202
Target Import	Target Import Set [imp_sn_gf_goal_target]	Target [sn_gf_goal_target]	true	100	true	202
Okta Example	label [oidc_okta_import]	User [sys_user]	true	100	true	201
Goal Import	Goal Import Set [imp_sn_gf_goal]	Goal [sn_gf_goal]	true	100	true	202
LDAP User Import	label [ldap_import]	User [sys_user]	true	100	true	201
LDAP Group Import	label [ldap_group_import]	Group [sys_user_group]	true	100	true	200
employee transform	import employee [u_import_employee]	User [sys_user]	true	100	true	202

1 to 15 of 15

servicenow All Favorites History Workspaces Table Transform Map - employee transform Search

Table Transform Map
employee transform

Copy Update

Enforce mandatory fields: No
Copy empty fields: ☐
Create new record on empty coalesce fields: ☐

Copy Update Delete

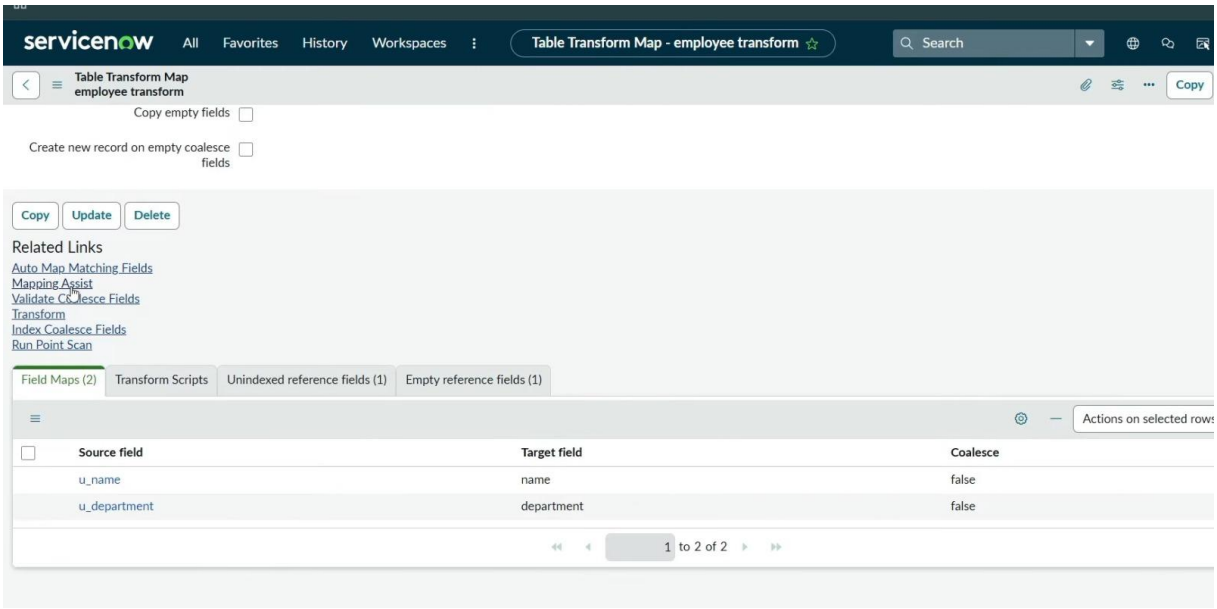
Related Links
Auto Map Matching Fields
Mapping Assist
Validate Coalesce Fields
Transform
Index Coalesce Fields
Run Point Scan

Field Maps (2) Transform Scripts Unindexed reference fields (1) Empty reference fields (1)

Actions on selected ro

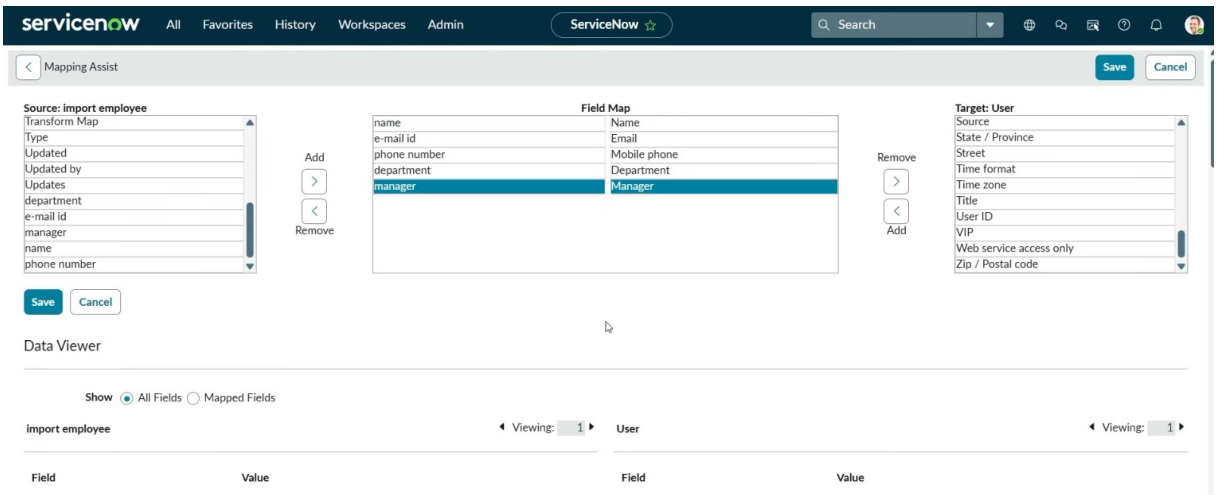
Source field	Target field	Coalesce
u_name	name	false
u_department	department	false

1 to 2 of 2



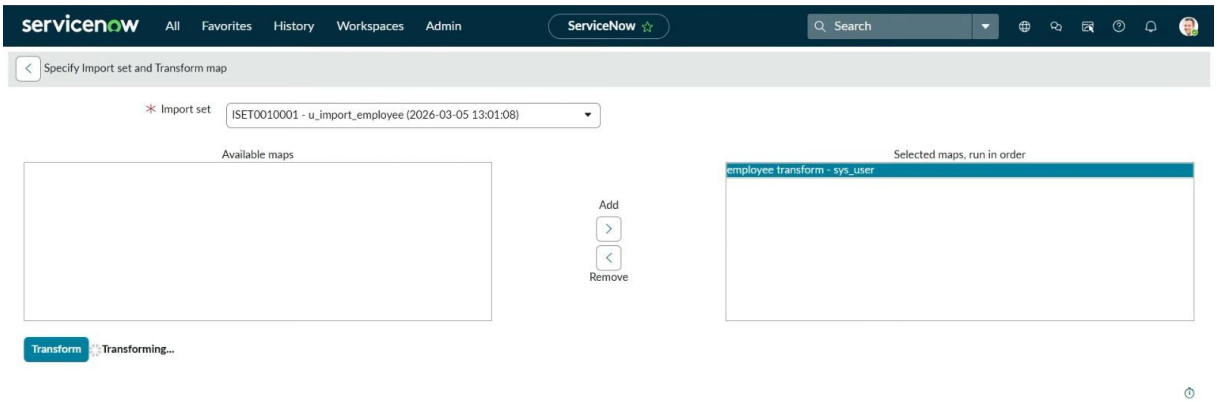
Step 4: Field Mapping

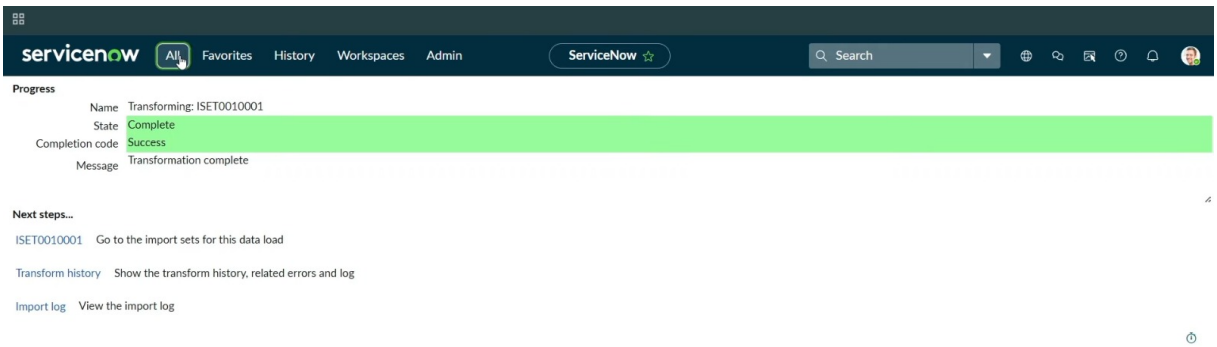
- Use Auto Match Fields for common mappings
- Use Mapping Assist for unmatched fields
- Map:
    - Manager (Reference field)
    - Department (Reference field)



Step 5: Execute Transform

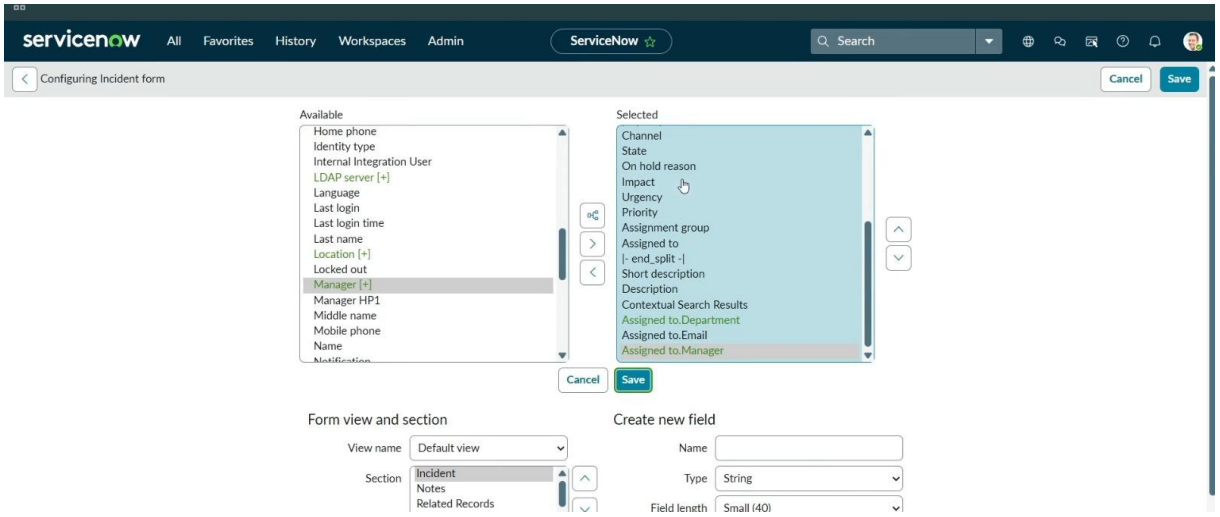
- Click on Transform
- Monitor transform state
- Verify inserted records





Step 6: Implement Dot Walking

- Navigate to Incident form
- Configure form layout
- Add fields:
    - Assigned To.Department
    - Assigned To.Email
    - Assigned To.Manager



Implementation:

The implementation phase translates the methodology into practical execution.

1. Data Creation

An employee dataset was created in Excel format with necessary fields.

2. Import Set Creation

The dataset was uploaded into ServiceNow using the Load Data module. A staging table was automatically created to store the data.

3. Transform Map Configuration

A Transform Map named "Test" was created. It connected the import set table with the User table.

4. Field Mapping

- Auto mapping matched similar fields
- Mapping Assist was used for:
  - Manager
  - Department

This ensured proper relationship mapping between records.

5. Data Transformation

The transform process was executed successfully. The system inserted multiple records into the User table.

6. Dot Walking Configuration

The Incident form was customized:

- Added department, email, and manager fields
- Used dot walking to retrieve related data

7. Testing

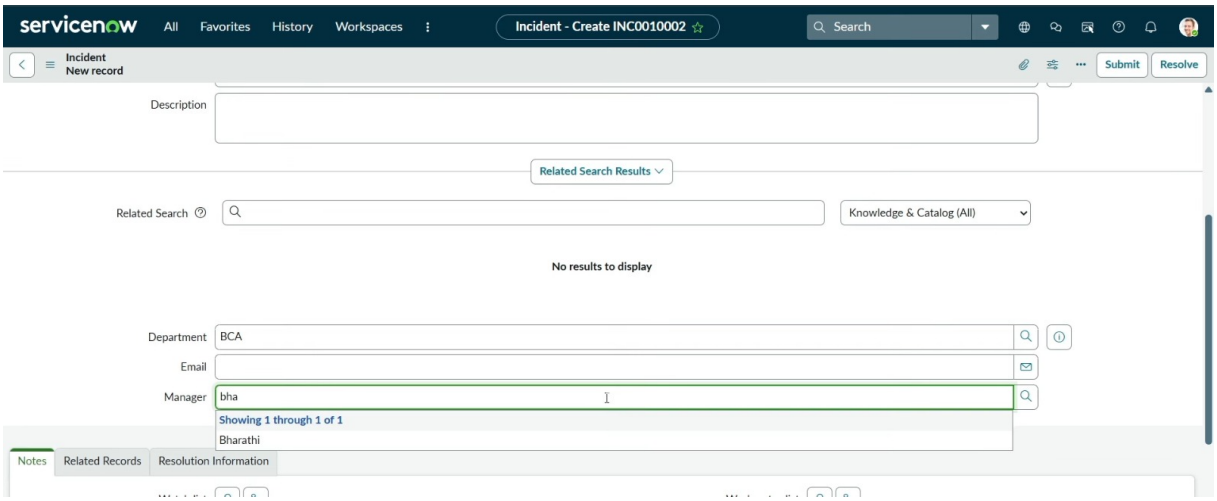
- Created a new incident
- Assigned it to an imported user
- Observed automatic population of related fields

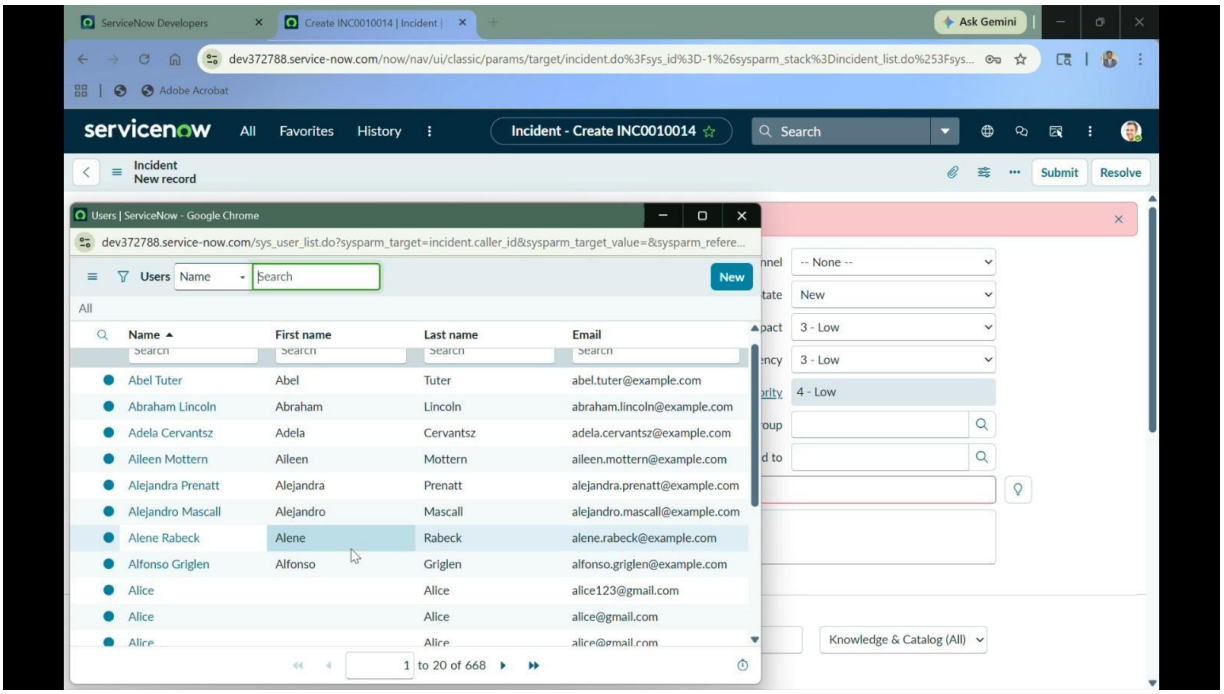
Results:

The project produced the following results:

- Successful import of employee data into ServiceNow
- Accurate mapping of relationships between users, managers, and departments

- Automatic retrieval of related data using dot walking
- Reduced manual effort and improved efficiency





Key Outcomes

- High data accuracy
- Improved user experience
- Efficient data handling
- Reduced redundancy

Conclusion:

This project demonstrates how ServiceNow's Import Sets and Transform Maps can be used to automate data import processes effectively. By properly mapping fields and maintaining relationships, the system ensures accurate and structured data storage. The use of dot walking further enhances usability by allowing users to access related data without navigating multiple records. This significantly improves productivity and reduces time consumption.

Overall, the project provides a scalable and efficient solution for enterprise data management.

Future Scope:

The project can be enhanced in the following ways:

- Integration with external systems using REST APIs
- Scheduled data imports for automation
- Advanced validation and error handling
- Use of scripting in Transform Maps
- Dashboard creation for data visualization
- Expansion to other modules like CMDB and Asset Management

References:

- ServiceNow Documentation – Import Sets and Transform Maps
- ServiceNow Developer Portal
- https://www.youtube.com/watch?v=Hf63Px29Jho
- Online technical blogs and tutorials on ServiceNow
- ITSM (IT Service Management) concepts and practices